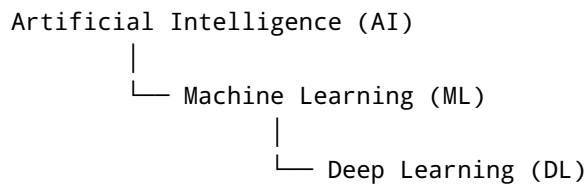


# AI vs ML vs Deep Learning

AI, Machine Learning, and Deep Learning are closely related, but they are **not the same thing**.

The easiest way to remember their relationship is:



**Deep Learning  $\subset$  Machine Learning  $\subset$  Artificial Intelligence**

## 1. Artificial Intelligence (AI)

### Definition

**Artificial Intelligence (AI)** is the broad field of creating computer systems that can perform tasks that normally require human intelligence.

These tasks can include:

- Reasoning
- Problem-solving
- Decision-making
- Understanding language
- Recognizing images
- Planning
- Learning

### Simple Example

A chess-playing AI can:

```
Look at the chess board
↓
Analyze possible moves
↓
Choose the best move
```

The overall goal is to make the system behave intelligently.

### Other examples

- Voice assistants
- Chess-playing systems
- Recommendation systems
- Autonomous vehicles
- Chatbots
- Medical diagnosis systems

---

## 2. Machine Learning (ML)

### Definition

**Machine Learning is a subset of AI that allows computers to learn patterns from data and use those patterns to make predictions or decisions.**

Instead of programming every rule manually, we provide data and allow the algorithm to learn.

### Example: House Price Prediction

Suppose we have:

| Size       | Bedrooms | Price |
|------------|----------|-------|
| 1000 sq ft | 2        | ₹50 L |
| 1500 sq ft | 3        | ₹70 L |
| 2000 sq ft | 4        | ₹95 L |

The ML algorithm learns the relationship:

```
House features
  ↓
ML algorithm
  ↓
Learned pattern
  ↓
Predicted price
```

For a new 1800 sq ft house, the model might predict its price.

## Common ML algorithms

- Linear Regression
  - Logistic Regression
  - KNN
  - Decision Tree
  - Random Forest
  - SVM
  - XGBoost
  - K-Means
- 

## 3. Deep Learning (DL)

### Definition

**Deep Learning is a subset of Machine Learning that uses artificial neural networks with multiple layers to learn complex patterns from data.**

Deep Learning is especially useful for:

- Images
- Video
- Speech
- Natural language
- Large amounts of unstructured data

### Example: Cat vs Dog Recognition

We give a neural network thousands of images:

```
Images
  ↓
Neural Network
  ↓
Learn patterns
  ↓
Cat / Dog
```

The network can learn useful visual features automatically.

Conceptually:

```

Image
↓
Input Layer
↓
Hidden Layer
↓
Hidden Layer
↓
Hidden Layer
↓
Output Layer
↓
Cat / Dog

```

## 4. AI vs ML vs DL

|                     | AI                                 | ML                     | DL                                  |
|---------------------|------------------------------------|------------------------|-------------------------------------|
| Full form           | Artificial Intelligence            | Machine Learning       | Deep Learning                       |
| Meaning             | Broad field of intelligent systems | Learning from data     | Learning using deep neural networks |
| Relationship        | Broadest                           | Subset of AI           | Subset of ML                        |
| Data                | May or may not require data        | Usually requires data  | Usually requires lots of data       |
| Feature engineering | Depends on approach                | Often important        | Often learned automatically         |
| Computing power     | Depends                            | Moderate to high       | Often high                          |
| Example             | Chess AI                           | House price prediction | Image recognition                   |

## 5. The Difference Using One Example

Imagine we want to build an **email spam detector**.

### AI

The goal is:

Build an intelligent system that can identify spam.

## ML

We provide thousands of emails labeled:

```
Email 1 → Spam  
Email 2 → Not Spam  
Email 3 → Spam
```

The ML algorithm learns patterns and predicts whether new emails are spam.

## DL

Instead of using traditional ML features, we could use a deep neural network that learns complex representations from the email text.

So:

```
AI  
↓  
Goal: Intelligent spam detection  
  
ML  
↓  
Learn patterns from email data  
  
DL  
↓  
Use deep neural networks to learn complex patterns
```

---

## 6. Traditional Programming vs ML

### Traditional Programming

We manually provide the rules:

```
Rules + Data  
↓  
Program
```

↓  
Output

Example:

```
if age >= 18:  
    print("Adult")  
else:  
    print("Minor")
```

---

## Machine Learning

We provide examples:

```
Data + Answers  
    ↓  
ML Algorithm  
    ↓  
Model  
    ↓  
Prediction
```

The model learns the relationship from the examples.

---

## 7. AI Does NOT Always Mean Machine Learning

This is an important point.

AI can be created using different techniques.

For example, a simple rule-based system:

```
if message == "hello":  
    response = "Hello!"
```

This can be part of an AI system, but it isn't Machine Learning because the computer **didn't learn the rule from data**.

AI can include:

- Rule-based systems
- Expert systems
- Search algorithms
- Planning
- Machine Learning
- Deep Learning

---

## 8. Where Does Generative AI Fit?

Generative AI is **not exactly a separate level above AI**.

Modern Generative AI commonly uses deep learning models, especially **transformers**.

For example:

```
Artificial Intelligence
      ↓
Machine Learning
      ↓
Deep Learning
      ↓
Neural Networks
      ↓
Transformers
      ↓
Generative AI applications
```

Generative AI can be used to generate:

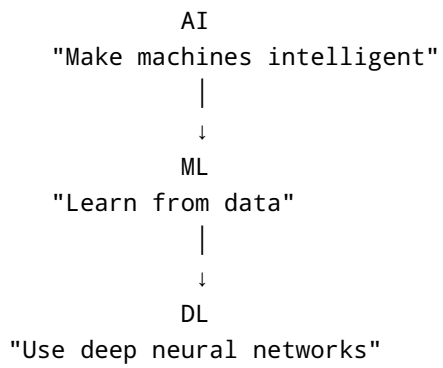
- Text
- Images
- Audio
- Video
- Code

Examples include AI chatbots, image generators, and code-generation systems.

---

## 9. Easy Real-Life Analogy

Think of it as a family:



**AI = The big field**

**ML = Learning from data**

**DL = Learning with deep neural networks**

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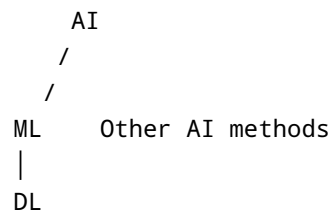
## 10. Easy Way to Remember

🧠 AI → Intelligence

📊 ML → Learning from Data

🕸 DL → Deep Neural Networks

And remember:



**So not all AI is ML, and not all ML is Deep Learning.**

---



## 11. Real-World Examples

| Application               | AI | ML        | DL            |
|---------------------------|----|-----------|---------------|
| Chess-playing system      | ✓  | Sometimes | Sometimes     |
| House price prediction    | ✓  | ✓         | Not necessary |
| Customer churn prediction | ✓  | ✓         | Not necessary |
| Customer segmentation     | ✓  | ✓         | Not necessary |
| Image recognition         | ✓  | ✓         | ✓             |
| Speech recognition        | ✓  | ✓         | ✓             |
| Large language models     | ✓  | ✓         | ✓             |
| Generative AI             | ✓  | ✓         | ✓             |

## 12. Interview Questions

### Q1. Is Machine Learning a part of AI?

**Yes.** Machine Learning is a subset of Artificial Intelligence.

### Q2. Is Deep Learning a part of Machine Learning?

**Yes.** Deep Learning is a subset of Machine Learning.

### Q3. Is every AI system a Machine Learning system?

**No.**

AI can also use rules, search, planning, expert systems, and other techniques.

### Q4. Is every Machine Learning model Deep Learning?

**No.**

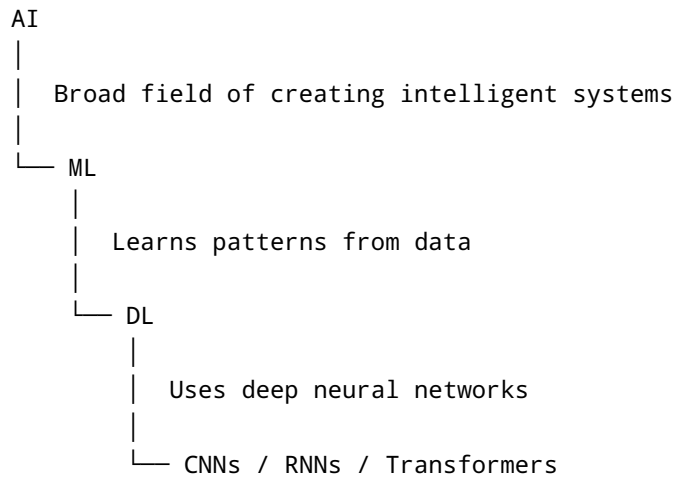
Linear Regression, Random Forest, XGBoost, KNN, and SVM are examples of ML techniques that aren't Deep Learning.

### Q5. Why is Deep Learning called "deep"?

Because it uses neural networks with **multiple layers** that can learn increasingly complex representations.



## Final Summary



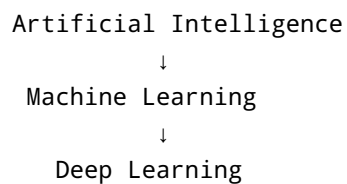
### One-line definitions

**AI:** Making machines perform tasks that require intelligence.

**ML:** Teaching machines to learn patterns from data.

**DL:** Using multi-layer neural networks to learn complex patterns.

### ★ Most Important Relationship



**AI is the broadest concept → ML is a subset of AI → DL is a subset of ML.**